

Summary of SVD

Given $A \in \mathbb{C}^{m \times n}$ there exists unique orthonormal bases $\{\vec{u}_1, \dots, \vec{u}_m\}$ of $\mathbb{C}^m$ and $\{\vec{v}_1, \dots, \vec{v}_n\}$ of $\mathbb{C}^n$ and singular values $\sigma_1, \dots, \sigma_r \in \mathbb{R}_+$ such that

$$\bullet \quad A \vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & 1 \leq i \leq r \\ \vec{0} & \text{else} \end{cases}$$

$$\bullet \quad \{\vec{v}_1, \dots, \vec{v}_n\} \text{ are e-vectors of } A^* A$$

$$A^* A \vec{v}_i = \begin{cases} \sigma_i^2 \vec{v}_i & 1 \leq i \leq r \\ 0 & \text{else} \end{cases}$$

$$\bullet \quad \{\vec{u}_1, \dots, \vec{u}_m\} \text{ are e-vectors of } A A^*$$

$$A A^* \vec{u}_i = \begin{cases} \sigma_i^2 \vec{u}_i & 1 \leq i \leq r \\ \vec{0} & \text{else} \end{cases}$$

$$\bullet \quad A = U \Sigma V^* \quad \text{where} \quad U = [\vec{u}_1 \dots \vec{u}_m],$$

$$V = [\vec{v}_1 \dots \vec{v}_n], \quad \text{and} \quad \Sigma = \begin{bmatrix} \sigma_1 & \dots & \sigma_r \\ & & \end{bmatrix}_{m \times n}$$

Pseudo - Inverse

Given A as above, define A^+ by

$$A^+ \vec{u}_i = \begin{cases} \sigma_i^{-1} \vec{v}_i & 1 \leq i \leq r \\ \vec{0} & \text{else} \end{cases}$$

or $A^+ = V \Sigma^+ U^*$ where $\Sigma^+ = \begin{bmatrix} \sigma_1^{-1} & & \\ & \ddots & \\ & & \sigma_r^{-1} \end{bmatrix}_{n \times m}$
reciprocal transpose

- AA^+ and A^+A are the orthogonal projections onto range A and range A^* respectively

Applications of SVD

- PCA (Principal Component Analysis)
- Machine Learning (in particular facial recognition)

Exercise:

- 1) Show that AA^+ is the orthogonal projection onto the range A .

We will show that

$$i. (AA^+)^2 = AA^+$$

$$ii. (AA^+)^* = AA^+$$

$$iii. AA^+|_{\text{range } A} = I$$

$$i. (AA^+)^2 = AA^+$$

Setting $A = U\Sigma V^*$ so that $A^+ = V\Sigma^+U^*$

we have that

$$(AA^+)^2 = (AA^+)(AA^+)$$

$$= (U\Sigma\Sigma^+U^*)(U\Sigma\Sigma^+U^*)$$

$$= U(\Sigma\Sigma^+)(\Sigma\Sigma^+)U^*$$

$$= U\Sigma\Sigma^+U^* = AA^+$$

$$U^{-1} = U^*$$

$$V^{-1} = V^*$$

$$\Sigma\Sigma^+ = \begin{bmatrix} I_{r \times r} & \\ & 0 \end{bmatrix}$$

orthogonal projection
on $\mathbb{R}^r$

$$ii. (AA^+)^* = AA^+$$

$$(AA^+)^* = (U\Sigma\Sigma^+U^*)^*$$

$$= U(\Sigma\Sigma^+)^*U^*$$

$$= U\Sigma\Sigma^+U^* = AA^+$$

$$\text{iii. } AA^+|_{\text{range } A} = I$$

$$\text{let } A \vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & 1 \leq i \leq r \\ \vec{0} & \text{else} \end{cases}$$

↗ rank A

$$\text{so that } A^+ \vec{u}_i = \begin{cases} \sigma_i^{-1} \vec{v}_i & 1 \leq i \leq r \\ \vec{0} & \text{else} \end{cases}.$$

Now let $\vec{x} \in \text{range } A$ and set

$$\vec{x} = \sum_{i=1}^r \alpha_i \vec{u}_i. \quad \text{Then as}$$

$$AA^+ \vec{x} = AA^+ \left(\sum_{i=1}^r \alpha_i \vec{u}_i \right)$$

$$= A \left(\sum_{i=1}^r \alpha_i A^+ \vec{u}_i \right)$$

$$= A \left(\sum_{i=1}^r \alpha_i \sigma_i^{-1} \vec{v}_i \right)$$

$$= \sum_{i=1}^r \alpha_i \sigma_i^{-1} A \vec{v}_i$$

$$= \sum_{i=1}^r \alpha_i \sigma_i^{-1} \sigma_i \vec{u}_i$$

$$= \sum_{i=1}^r \alpha_i \vec{u}_i = \vec{x}$$

For $\vec{x} \in \mathbb{R}^m$ s.t.

$$\vec{x} = \sum_{i=1}^m \alpha_i \vec{u}_i,$$

$$AA^+ \vec{x} = \sum_{i=1}^r \alpha_i \vec{u}_i$$

Matrix Norm

Def 23.1 Let V be a vector space. Then

$\|\cdot\| : V \rightarrow [0, \infty)$ is a vector norm iff

- i. $\|\vec{0}\| = 0$
- ii. $\|\vec{v}\| > 0$ for $\vec{v} \neq \vec{0}$
- iii. $\|\lambda \vec{v}\| = |\lambda| \cdot \|\vec{v}\|$ for any $\lambda \in F$
- iv. (Triangle inequality) $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$

Topological spaces $\supset$ Metric space $\supset$ normed
vector spaces $\supset$ inner product spaces

Def 23.2 Let $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}_{\geq 0}$ be defined

$$\text{as } \|A\| = \max_{\vec{x} \neq \vec{0}} \frac{\|A\vec{x}\|}{\|\vec{x}\|} = \max_{\|\vec{x}\|=1} \|A\vec{x}\|$$

$\swarrow$
norm of A

$$\|A\vec{x}\| \leq \|A\| \cdot \|\vec{x}\|$$

Theorem 23.3 $\|\cdot\|$ is norm on $\mathbb{C}^{m \times n}$ for any
 $m, n \in \mathbb{N}$.

Proof: i, ii. are clear (note that if $A \neq 0$, then $A\vec{x} \neq \vec{0}$ for some $\vec{x} \neq \vec{0}$)

iii. Also clear as $\|\lambda A\vec{x}\| = |\lambda| \|A\vec{x}\|$

iv. Follows from the triangle inequality

$$\text{on } \mathbb{C}^n: \| (A+B)\vec{x} \| = \| A\vec{x} + B\vec{x} \| \\ \leq \| A\vec{x} \| + \| B\vec{x} \|$$

Theorem 23.4 (Cauchy-Schwarz) For $A, B \in \mathbb{C}^{n \times n}$

$$\|AB\| \leq \|A\| \cdot \|B\|$$

(compare with $|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\|$ in $\mathbb{C}^n$)

Proof: As $\|A\| = \max_{\|\vec{x}\|=1} \|A\vec{x}\|$ we have that

$$\|AB\vec{x}\| = \|A(B\vec{x})\| \leq \|A\| \cdot \|B\vec{x}\| \\ \leq \|A\| \cdot \|B\| \cdot \|\vec{x}\|$$

for all $\vec{x} \in \mathbb{C}^n$.

Now similarly we define the operator norm on $\mathcal{L}(\mathbb{C}^n, \mathbb{C}^m)$ as

$$\|T\| = \max_{\vec{x} \neq \vec{0}} \frac{\|T\vec{x}\|}{\|\vec{x}\|}$$

which satisfies the same properties as above

i. $\|T\|$ is pos. def.

ii. $\|\lambda T\| = |\lambda| \cdot \|T\|$

iii. $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$

iv. $\|T_1 \circ T_2\| \leq \|T_1\| \cdot \|T_2\|$

Now let $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ and let $A = [T]$
so that $A \in \mathbb{C}^{n \times n}$.

Theorem 23.5 let A have singular values of
 $\sigma_1, \dots, \sigma_r$. Then $\|T\| = \|A\| = \max \sigma_i$
 $= \max \sigma(A)$

$\|A\|$ is the max singular value

Proof: Since $\frac{\|A\vec{x}\|^2}{\|\vec{x}\|^2} = \frac{\vec{x}^*(A^*A)\vec{x}}{\vec{x}^*\vec{x}}$ is

bounded by the e-values of A^*A

(see HW 8 #5) which are σ_i^2 we

have that $\frac{\|A\vec{x}\|}{\|\vec{x}\|} \leq \max \sigma_i$.

Def 23.6 $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^m)$ is bounded iff
for $\vec{x} \in \mathbb{C}^n$, there exists $C > 0$ s.t. $\|T\vec{x}\| \leq C \|\vec{x}\|$
↓
 $\|T\|$

Theorem 23.7 T is bounded and thus
continuous. All (finite dim) matrices are continuous

pick $\delta = \frac{\epsilon}{\|T\|}$

see polar decomp.

Theorem 23.8 Given $A = U\Sigma V^*$ where U and
 V are unitary, $Q = UV^*$ is the "closest"
unitary matrix to A in matrix norm. I.e.

$$Q = \min_{Q \text{ unitary}} \|A - Q\|$$